

Article

Comparative Analysis of Resampling Techniques for Class Imbalance in Financial Distress Prediction Using XGBoost

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Abstract

One of the key challenges in financial distress data is class imbalance, where the data are characterized by a highly imbalanced ratio between the number of distressed and non-distressed samples. This study examines eight resampling techniques for improving distress prediction using the XGBoost algorithm. The study was performed on a dataset acquired from the CSMAR database, containing 26,383 firm-quarter samples from 639 Chinese A-share listed companies (2007–2024), with only 12.1% of the cases being distressed. Results show that standard Synthetic Minority Oversampling Technique (SMOTE) enhanced F1-score (up to 0.73) and Matthews Correlation Coefficient (MCC, up to 0.70), while SMOTE-Tomek and Borderline-SMOTE further boosted recall, slightly sacrificing precision. These oversampling and hybrid methods also maintained reasonable computational efficiency. However, Random Undersampling (RUS), though yielding high recall (0.85), suffered from low precision (0.46) and weaker generalization, but was the fastest method. Among all techniques, Bagging-SMOTE achieved balanced performance (AUC 0.96, F1 0.72, PR-AUC 0.80, MCC 0.68) using a minority-to-majority ratio of 0.15, demonstrating that ensemble-based resampling can improve robustness with minimal impact on the original class distribution, albeit with higher computational cost. The compared findings highlight that no single approach fits all use cases, and technique selection should align with specific goals. Techniques favoring recall (e.g., Bagging-SMOTE, SMOTE-Tomek) are suited for early warning, while conservative techniques (e.g., Tomek Links) help reduce false positives in risk-sensitive applications, and efficient methods such as RUS are preferable when computational speed is a priority.

Keywords: class imbalance; SMOTE; ADASYN; Borderline-SMOTE; SMOTE-Tomek; SMOTE-ENN; RUS; Tomek Links; Bagging-SMOTE; XGBoost

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1. Introduction

In recent years, the application of machine learning to financial distress prediction has attracted increasing attention, owing to its powerful capabilities in feature extraction and nonlinear modeling, which significantly enhance the ability to capture complex patterns in financial data and improve predictive accuracy. However, a core challenge remains: the severe class imbalance inherent in financial datasets. In practice, distressed samples constitute only a small fraction of all listed companies, causing models to be biased toward

the majority class (non-distressed samples) during training. This often results in high overall accuracy but poor recall for the minority class (distressed samples), which greatly undermines the practical value of predictive systems in financial risk management.

Under such circumstances, classification metrics such as accuracy, recall, and precision often fail to reflect a model's true ability to identify distressed samples. Therefore, effectively addressing the class imbalance problem has become a critical issue in financial distress prediction. Resampling techniques are among the most widely adopted solutions, as they adjust the distribution of training samples to achieve class balance. Commonly used oversampling techniques include Synthetic Minority Oversampling Technique (SMOTE), Adaptive Synthetic Sampling (ADASYN), and Borderline-SMOTE, as well as hybrid approaches such as SMOTE-Tomek and SMOTE-ENN. Meanwhile, undersampling techniques such as Random Undersampling (RUS) and Tomek Links are also frequently used to reduce the dominance of the majority class. While these resampling techniques have shown advantages in various imbalanced classification tasks, there is still a lack of comprehensive comparative studies that assess their effectiveness and applicability in enhancing model performance for real-world financial distress detection.

Despite the growing body of research on resampling techniques, most existing studies either focus on a limited subset of techniques or rely on simulated or non-financial datasets, which may not fully capture the unique characteristics and challenges of real-world financial data. Furthermore, previous comparative analyses often lack a unified experimental framework, making it difficult to draw consistent and practical conclusions regarding the relative performance of different resampling techniques in financial distress prediction.

To address these gaps, this study presents a systematic and in-depth comparative analysis of a broad range of resampling techniques, including SMOTE, ADASYN, Borderline-SMOTE, SMOTE-Tomek, SMOTE-ENN, Random Undersampling, Tomek Links, and Bagging-SMOTE, using real financial data from publicly listed companies. All methods are evaluated under consistent experimental conditions, with XGBoost employed as the base classifier and a unified evaluation framework applied throughout. This approach enables a fair and comprehensive assessment of the strengths and limitations of each technique in the context of financial distress prediction.

By comparing the performance of these methods across multiple metrics relevant to imbalanced classification such as F1-score, area under the receiver operating characteristic curve (AUC), area under the precision-recall curve (PR-AUC), and Matthews correlation coefficient (MCC), we reveal the effectiveness of different resampling techniques in enhancing the identification of distressed samples and provide practical, evidence-based guidance for practitioners and researchers in the field of financial risk management.

2. Related Works

Class imbalance is a common challenge when working with real-world data, often leading to degraded classification performance. This is because most machine learning algorithms are not inherently designed to handle highly imbalanced datasets effectively. Machine learning models favor the majority class, thereby reducing the accuracy of predictions for the minority class. Song et. al. [1] and Engin [2] highlighted this issue and employed various resampling techniques, including SMOTE and ensemble methods, to reshape the dataset size to achieve class balance, thereby improving model performance.

Resampling techniques can generally be categorized into three major groups: oversampling, undersampling, and hybrid resampling. Each approach seeks to address class imbalance by adjusting the size of the majority or minority class to enhance the model's prediction power in detecting minority class instances.

2.1. Oversampling Techniques

Oversampling techniques address class imbalance by generating synthetic samples for the minority class, thereby improving the performance of machine learning models in detecting rare but critical events.

The commonly used oversampling technique is the Synthetic Minority Oversampling Technique (SMOTE), which generates synthetic samples for the minority class by interpolating new synthetic instances between minority class instances and their nearest neighbors [3]. It has been widely applied in biomedical domains to generate synthetic samples of the minority class in protein-protein interactions [4], breast cancer prediction [5], and medical diagnosis [6]. In some practice studies, Alex and Nayahi [7] reported that SMOTE introduced noise during the process of creating synthetic samples. This could be due to instances that are sparsely distributed, and synthetic samples do not accurately reflect the true characteristics of the minority class.

To alleviate the limitations of SMOTE, Han, Wang, and Mao [8] proposed using Borderline-SMOTE, which focuses on generating samples near the class boundary rather than the entire minority class. It operates on the premise that minority samples far from the class boundary contribute minimally to classification improvement, while those closer to the boundary play a critical role in distinguishing between classes. The effectiveness of Borderline-SMOTE has been reported in fault detection systems [9] and text classification [10], where careful parameter tuning is crucial to prevent misclassification at the class boundaries.

Adaptive Synthetic Sampling (ADASYN) is an advanced oversampling technique that extends SMOTE by adaptively focusing on minority class instances that are harder to learn [11]. The method increases the sampling rate for minority instances that are misclassified or located near the decision boundary, thereby enhancing the classifier's focus on complex regions of the input space. It has been reported to outperform other methods in multiclass imbalanced scenarios when combined with stacking algorithms [12]. Analogous to other oversampling techniques, ADASYN can easily overfit noisy regions if not combined with any other denoising techniques.

2.2. Undersampling Techniques

Unlike oversampling techniques that increase the number of instances for the minority class, undersampling techniques tackle class imbalance by reducing the number of instances in the majority class.

Random Under Sampling (RUS) is an undersampling technique that reduces the size of the majority class by randomly removing instances. Due to its random nature, RUS carries the risk of discarding important information, which is why it is often combined with cleaning strategies. Arifin et al. [13] showed that RUS with decision tree and random forest classifiers performed better than other resampling techniques.

Tomek Links is another widely used undersampling technique, which identifies minority-majority pairs where each is the nearest neighbor of the other, and removing these pairs improves class separation [14]. This technique has been applied in credit card fraud detection [15] and abnormal traffic detection for filtering noise samples in the data [16]. However, it may excessively trim borderline instances to cause an overly aggressive reduction in the dataset's complexity, potentially losing valuable, nuanced patterns that could contribute to a more comprehensive understanding of the predictive model's behavior and performance.

2.3. Hybrid Resampling Techniques

Although oversampling and undersampling techniques are effective for balancing data, relying on a single resampling method has its limitations. To address these shortcomings, hybrid techniques that combine multiple resampling approaches have been proposed.

SMOTE-Tomek, which combines oversampling and undersampling techniques, is widely used in enhancing classifier performance for cancer prediction datasets [17]. It combines SMOTE with Tomek Links to remove noisy majority samples near minority instances, which needed careful validation to avoid overfitting the cleaned subsets.

SMOTE-ENN integrates SMOTE with Edited Nearest Neighbors (ENN) to delete misclassified majority samples post-oversampling, refining decision boundaries. It has been used in medical datasets such as missed abortion diagnosis to enhance the model's classification accuracy [18]. Similar to most of the undersampling techniques, ENN may remove useful samples in its aggressive pruning process.

2.4. Prediction Models with Built-In Resampling Techniques

A major challenge in hybrid resampling techniques is maintaining a balance between enhancing minority class representation and preserving the overall data quality. This enhancement is typically achieved by reducing the number of majority class samples and/or increasing instances of the minority class. However, excessive pruning or over-generating samples can distort the original data distribution, potentially leading to inaccurate predictions.

Recent studies have demonstrated the effectiveness of XGBoost in handling the class imbalance issue of financial distress scenarios [19,20]. XGBoost incorporates cost-sensitive learning through adjustable weights for minority class instances, optimizing prediction. It employs regularization techniques to prevent overfitting, making it suitable for high-dimensional financial datasets [21].

RUSBoost, which combines Random Undersampling (RUS) with boosting algorithms, was also explored in various application scenarios [22]. It reduces the size of the majority class to balance the dataset, mitigating bias toward majority classes and boosting iteratively trained weak classifiers, emphasizing misclassified instances in subsequent iterations.

2.5. Summary

In summary, oversampling techniques such as SMOTE and ADASYN have proven effective in improving minority class detection, particularly in dense regions of the minority class [23]. However, in financial distress prediction, where distressed companies may be highly heterogeneous and sparsely distributed, these methods risk introducing synthetic noise or generating unrealistic samples [24], potentially distorting the true risk profile of companies. Undersampling techniques such as Tomek Links can enhance class separation and reduce overlap, but they may also discard valuable information from the majority class, which is especially problematic in financial datasets where each observation may carry unique, informative signals about corporate health.

Hybrid techniques, such as SMOTE-Tomek and SMOTE-ENN, often outperform individual methods by combining data synthesis with noise reduction [25]. Nevertheless, these approaches can lead to over-pruning, inadvertently removing borderline or informative samples along with noise, which may reduce the model's ability to capture subtle patterns critical for early warning in financial distress scenarios. Moreover, the effectiveness and side effects of these resampling techniques are highly dependent on dataset characteristics, such as the degree of imbalance, feature distribution, and the presence of outliers, as well as on computational resources, an important consideration for large-scale financial applications.

Previous comparative studies have often focused on general classification tasks or biomedical datasets, with limited attention to the unique challenges posed by financial distress prediction. These limitations underscore the need for a systematic evaluation of resampling techniques in the context of real-world financial datasets, considering both predictive performance and practical trade-offs.

This study aims to address the identified research gaps by critically evaluating the effectiveness and limitations of major resampling techniques on a large, real-world financial dataset. We utilize a comprehensive set of financial indicators, encompassing both general financial ratios and bankruptcy-related features, to ensure robust analysis. Additionally, our study considers computational efficiency and discusses the practical implications of different resampling techniques for financial risk management.

3. Materials and Methods

3.1. Data

The data used in this study were obtained from the China Stock Market & Accounting Research (CSMAR) database. It comprised quarterly financial data from 639 Chinese A-share listed companies from 2007 to 2024, resulting in a total of 26,383 firm-quarter samples with 368 variables. The data are highly imbalanced, with 3181 samples (12.1%) labeled as Special Treated (ST), and 23,202 samples (87.9%) labeled as non-ST (see Figure 1).

Table 1 summarizes the dataset, which includes two subsets: Financial Indicators and Bankruptcy Reorganization. Detailed descriptions are available in CSMAR. The financial indicators span various dimensions, while the bankruptcy reorganization subset adds variables related to business risk.

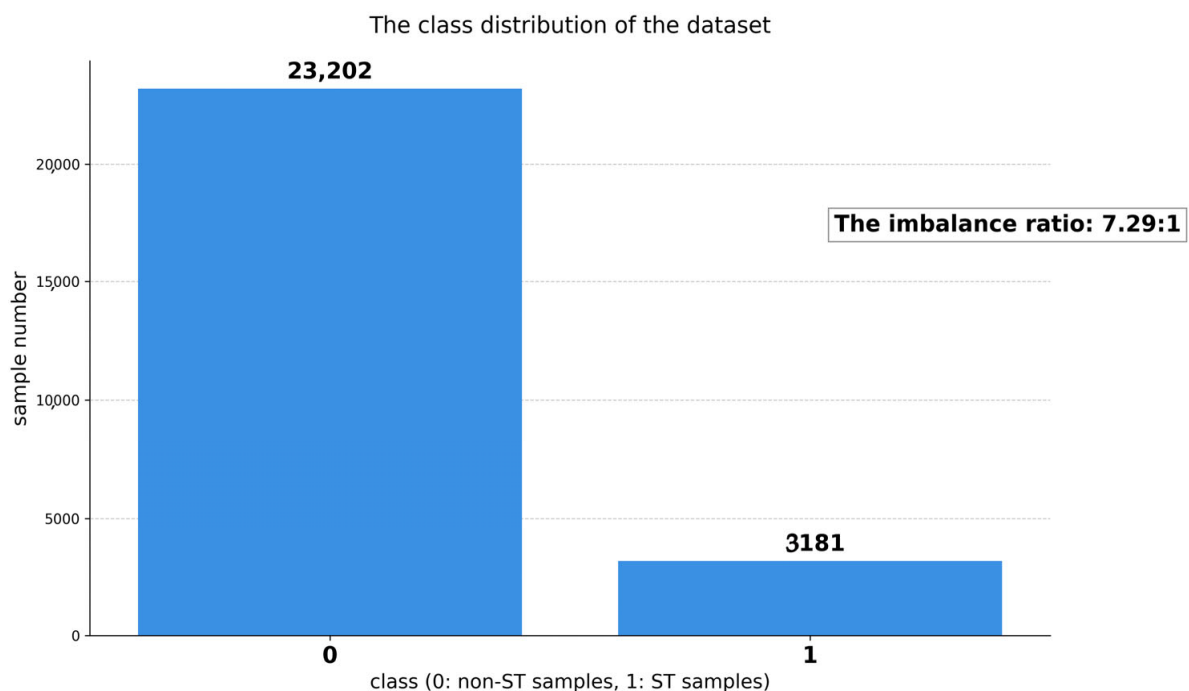


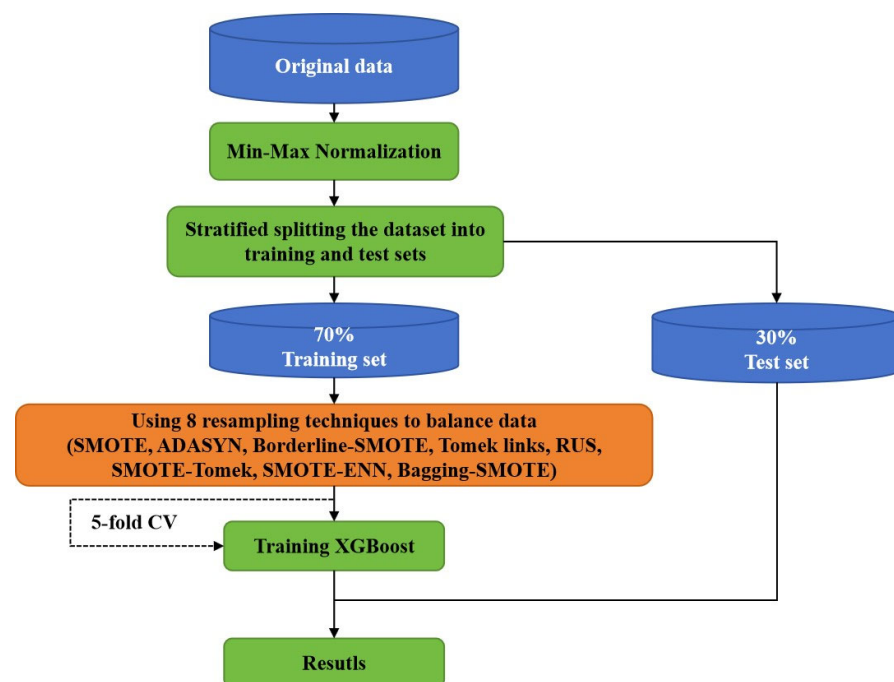
Figure 1. Sample distribution.

Table 1. The dataset components.

Set	Category	Brief Description
Financial Indicators	Solvency	Ability to repay short- and long-term debt using enterprise assets.
	Disclosed Index	Indicators directly reported to reflect company operations.
	Ratio Structure	Financial structure based on the proportions of key financial indicators.
	Operating Capacity	Efficiency in utilizing assets for business operations.
	Earning Capacity	Ability to generate profits.
	Cash Flow	Cash-related indicators derived from financial statement ratios.
	Risk Level	Risk of financial instability due to weak structure or poor financing practices.
	Growth Capability	Potential for future expansion and performance improvement.
	Index per Share	Financial condition measured on a per-share basis.
Bankruptcy Reorganization	Relative Value Index	Derived from comparisons among related indicators.
	Dividend Distribution	Metrics derived from comparisons among related financial indicators.
	Business Risk	Risk profile of bankrupt or restructured listed firms.

3.2. Experimental Design

Figure 2 illustrates the experimental workflow of this study. We first normalize the original dataset using min-max normalization. We then partition the data into training and test sets with a stratified split ratio of 70% for training and 30% for testing. To systematically investigate the impact of class imbalance handling, we compare eight resampling techniques, each applying exclusively to the training data to prevent data leakage. We then use each balanced training set to train the XGBoost model, employing 5-fold cross-validation to mitigate overfitting. Finally, we evaluate each trained XGBoost model on the untouched test set to assess generalization ability and robustness to unseen data.

**Figure 2.** Experimental workflow for the study.

3.2.1. Data Normalization

Min-max normalization was selected in this study to rescale feature values because it preserves the underlying distribution and patterns of the data. Additionally, since the base classifier used (XGBoost) does not assume normality, min-max scaling is a suitable choice. Equation (1) presents the mathematical expression of this method.

$$\text{Min-Max normalization, } X' = \frac{X_i - X_{\min}}{X_{\max} - X_{\min}} \quad (1)$$

Given a feature X consisting of value $\{X_1, X_2, \dots, X_i\}$, where i is the total number of observations in the feature, the normalization process involves subtracting the minimum value of X from each value and then dividing by the range. The resulting normalized values retain the original distribution shape but are confined within the range of $[0, 1]$.

3.2.2. Stratified Splitting of the Dataset

To ensure a fair and representative evaluation, we employ stratified splitting to divide the dataset into training and test sets. Stratified splitting preserves the original class distribution in both subsets, which is especially important for imbalanced classification tasks. Importantly, all subsequent data preprocessing steps, including resampling techniques such as oversampling, undersampling, and hybrid resampling, were performed exclusively on the training set. The test set remains completely isolated and was not used in any way during resampling or model training. This protocol effectively prevents data leakage, ensuring that the evaluation metrics reflect the model's true generalization ability on unseen data.

3.2.3. Resampling and Hybrid Techniques

(1) Synthetic Minority Oversampling Technique (SMOTE)

SMOTE is an oversampling technique that creates new instances through linear interpolation between a given minority sample and its k -nearest minority class neighbors. The default number of k neighbors in the standard SMOTE algorithm is 5.

The process involves selecting a minority class sample x_i , identifying one of its nearest neighbors x_j , and generating a synthetic sample x_{synth} as follows:

$$x_{\text{synth}} = x_i + \lambda \cdot (x_j - x_i) \quad (2)$$

where x_i , and x_j are minority class neighbors, and λ is a random weight $\in [0, 1]$. This operation effectively generates new samples along the line segments connecting minority instances, which helps to generalize the minority region and reduce model bias toward the majority class.

(2) Borderline-SMOTE

Borderline-SMOTE is a supervised oversampling method that improves upon the original SMOTE algorithm by focusing specifically on minority class instances that lie near the decision boundary, where misclassification is more likely. The technique starts by analyzing the k -nearest neighbors of each minority class instance. Based on the distribution of class labels within this neighborhood, each instance is categorized as safe, noisy, or dangerous. Only those identified as dangerous, which are surrounded by a mix of majority and minority class neighbors, are selected for oversampling.

For a given dangerous minority instance x_i and its k -nearest minority neighbors $\{k_1, k_2, \dots, k_j\}$, the SMOTE interpolation process generates synthetic samples by combining x_i with a selected neighbor k_j . The synthetic instance is created using a linear

interpolation between the two points based on their distance. The mathematical representation is given as follows:

$$x_{\text{synth}} = x_i + \lambda \cdot (x_j - x_i) \quad (3)$$

where λ is a random weight $\in [0, 1]$.

(3) Adaptive Synthetic Sampling (ADASYN)

ADASYN is an extension technique of SMOTE that focuses on minority class instances that are harder to learn. It begins by computing the ratio of majority class neighbors within the k -nearest neighborhood of each minority sample. A higher ratio indicates greater difficulty in learning that instance. ADASYN assigns a higher weight to such samples when generating synthetic data. Formally, for each minority sample x_i , the number of synthetic samples to be generated is proportional to its local density measure. Equation (4) is the mathematical expression of this method.

$$g_i = \frac{r_i}{\sum_{j=1}^n r_j} \quad (4)$$

where r_i is the ratio of the majority class examples to x_i , and n is the total number of observations in the minority class.

New instances are generated using the same interpolation mechanism as in SMOTE. By concentrating on more challenging areas of the feature space, ADASYN dynamically shifts the decision boundary and enhances the learning process.

(4) Random Undersampling (RUS)

RUS is an undersampling technique that reduces the number of majority class instances by randomly removing samples, thereby balancing the class distribution in the dataset.

The core assumption of RUS is that the majority class contains redundant or less informative instances that can be discarded without significantly compromising model performance. However, due to the random nature of instance removal, there is a potential risk of discarding important majority examples and thus losing valuable information.

(5) Tomek Links

Tomek Links is an undersampling technique that finds pairs of nearest-neighbor instances from different classes. When such pairs are found, instances of the majority class will be removed, thereby refining the decision boundary and reducing overlap between the two classes. Equation (5) shows the mathematical representation of this method.

$$NN(x_i) = x_j \wedge NN(x_j) = x_i \wedge y_i \neq y_j \quad (5)$$

Two instances (x_i, x_j) form a Tomek Link if they are nearest neighbors ($NN(x)$) from opposite classes (y_i, y_j) .

(6) SMOTE-Tomek

SMOTE-Tomek is a two-stage hybrid technique that integrates SMOTE oversampling with Tomek Links undersampling [26]. Firstly, SMOTE generates synthetic minority class samples using interpolation between existing instances and their k -nearest neighbors (see Equation (2)). In the latter stage, Tomek Links are identified within the augmented dataset to remove nearest-neighbor instances from the majority class (see Equation (5)).

The combination of oversampling through SMOTE and data cleaning via Tomek Links enables SMOTE-Tomek to enhance both class balance and the discriminative quality of the feature space.

(7) SMOTE-Edited Nearest Neighbors (SMOTE-ENN)

SMOTE-ENN is a hybrid resampling technique that integrates SMOTE with the ENN method [14]. This approach aims to address class imbalance and to enhance the quality of the training dataset by removing noisy or ambiguous instances.

In the first step, SMOTE generates synthetic instances for the minority class in the dataset. The ENN algorithm is then applied to remove any instance whose class label differs from the majority class of its k -nearest neighbors. This rule applies to both original and synthetic examples, enabling the algorithm to eliminate overlapping or noisy points from both classes. By combining data augmentation and noise filtering, SMOTE-ENN improves class separability and enhances generalization performance.

(8) Bagging-SMOTE

Bagging-SMOTE is a prediction model that combines the strengths of bootstrap aggregating and SMOTE to improve classification performance on imbalanced datasets. The training data were first randomly split into K bootstrap folds (with replacement). The SMOTE was then applied to each bootstrap fold to balance the class distribution. In the Bagging-SMOTE framework, we selected XGBoost as the base classifier.

Table 2 summarizes the resampling techniques used in this study.

Table 2. Summary of resampling and hybrid techniques.

Technique	Type	Description
SMOTE	Oversampling	Generates synthetic minority class samples via linear interpolation between neighbors.
ADASYN	Oversampling	Focuses synthesis on difficult minority samples by weighting based on local imbalance.
Borderline-SMOTE	Oversampling	Specifically, it oversamples minority instances near the decision boundary.
Random Under	Undersampling	Randomly reduces the majority class samples to balance the distribution.
Tomek Links	Undersampling	Removes overlapping majority class samples to improve class separation.
SMOTE-Tomek	Hybrid	Apply SMOTE, then clean the noise with Tomek Links.
SMOTE-ENN	Hybrid	Generates synthetic samples and then removes misclassified instances.
Bagging-SMOTE	Hybrid	Stratify data with sampling by replacement, then apply SMOTE in each stratified set of data.

3.2.4. XGBoost Prediction Model

XGBoost is a powerful gradient boosted trees algorithm known for its effectiveness in handling complex datasets. We select it as the base classifier in this study due to its strong performance in tabular data modeling, robustness to multicollinearity, and ability to handle missing values—characteristics well-suited to financial datasets with high dimensionality and sparsity. In addition, XGBoost offers effective regularization to prevent overfitting, supports various objective functions, and provides efficient, scalable implementations that are widely adopted in both academic research and industry. These advantages make it a

practical and reliable choice for our analysis focused on evaluating resampling techniques under class imbalance.

During model training, it begins with an initial prediction based on the distribution of the target variable. We then train the subsequent trees to correct the errors of the previous model by focusing on the misclassified samples. Each new tree works to refine the predictions of the previous model by re-predicting the incorrectly classified samples. This iterative process continues until either the predefined number of trees is built or the number of unchanged predictions reaches a specified threshold.

To mitigate overfitting, the experiment employs 5-fold cross-validation (CV) during training. In each iteration, it trains the model on four folds of the data, while the remaining fold is used to validate its predictive performance. This process is repeated until each fold has served once as the validation set. The experiment uses average performance across all folds to assess the model's generalization ability. Table 3 shows the used hyperparameters of the model.

Table 3. Hyperparameter in XGBoost.

Parameter	Value	Descriptions
random_state	42	To ensure the reproducibility of experimental results
n_estimators	300	The maximum number of trees
eval_metric	Logloss, AUC	To evaluate the performance of tree prediction
early_stopping_rounds	20	Early stopping to prevent model overfit

To comprehensively assess the predictive performance of the XGBoost classifier, we employ a range of evaluation metrics, including accuracy, precision, recall, F1-score, area under the receiver operating characteristic curve (AUC-ROC), area under the precision-recall curve (PR-AUC), and Matthews correlation coefficient (MCC). Equations (6)–(11) present the mathematical representation of these metrics.

$$\text{Accuracy} = \frac{\text{TP} + \text{TN}}{\text{TP} + \text{TN} + \text{FP} + \text{FN}} \quad (6)$$

$$\text{Precision} = \frac{\text{TP}}{\text{TP} + \text{FP}} \quad (7)$$

$$\text{Recall} = \frac{\text{TP}}{\text{TP} + \text{FN}} \quad (8)$$

where,

TP = True Positives

TN = True Negatives

FP = False Positives

FN = False Negatives

$$\text{F1 Score} = \frac{2 \times \text{Precision} \times \text{Recall}}{\text{Precision} + \text{Recall}} \quad (9)$$

$$\text{MCC} = \frac{(\text{TP} \times \text{TN}) - (\text{FP} \times \text{FN})}{\sqrt{(\text{TP} + \text{FP})(\text{TP} + \text{FN})(\text{TN} + \text{FP})(\text{TN} + \text{FN})}} \quad (10)$$

$$\text{AUC} = \int_0^1 \text{TPR}(\text{FPR})d(\text{FPR}) \quad (11)$$

where,

TPR (True Positive Rate) is also known as sensitivity or recall.

FPR (False Positive Rate) is defined as $\frac{FP}{FP+TN}$.

4. Results

To preserve the original class imbalance ratio, we stratified the dataset into training and test subsets. Figure 3 illustrates the class distribution of the entire dataset, training, and test subsets. The proportion of the minority class (i.e., Class 1, representing distressed samples) is consistently 12.1%, while the majority class (i.e., Class 0, non-distressed samples) accounts for 87.9% across all splits.

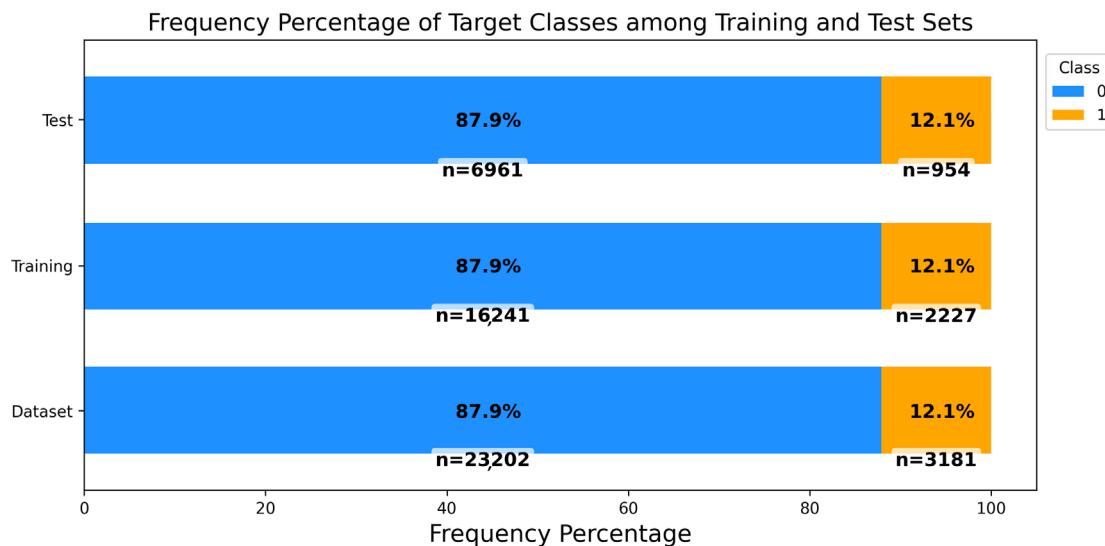


Figure 3. Class distribution among training and test sets.

4.1. Impact of Resampling Techniques Toward the Training Set

To address class imbalance, we applied eight resampling techniques to the training subset. Figure 4 compares the class distribution of the original training set against the effects of the eight resampling methods on the minority and majority classes.

SMOTE (see Figure 4b), ADASYN (see Figure 4c), and Borderline-SMOTE (see Figure 4d) generate synthetic minority samples to balance the dataset, with ADASYN focusing more on difficult-to-learn regions. SMOTE-Tomek (see Figure 4g) and SMOTE-ENN (see Figure 4h) combine oversampling with cleaning methods by removing noisy or overlapping majority class samples, resulting in clearer class boundaries. RUS (see Figure 4e) reduces the number of majority samples, leading to a more compact but potentially under-representative distribution. Tomek Links (see Figure 4f) remove borderline majority class samples that overlap or are close to the minority class, resulting in a cleaner class boundary. As shown by the purple '×' markers, a small subset of the majority class is eliminated, leading to minimal distributional shift while retaining the overall structure of the data. Bagging-SMOTE (see Figure 4i) produces a similarly distributed training set in each fold, following a sampling strategy of 0.15, where synthetic minority class samples are generated to reach approximately 15% of the majority class size. The synthetic minority samples (green triangles) enhance minority representation in overlapping areas of the feature space. These visualizations collectively demonstrate how each technique reshapes the data space to address class imbalance and enhance classifier learning.

Table 4 summarizes the class distribution comparison before and after applying the resampling techniques.

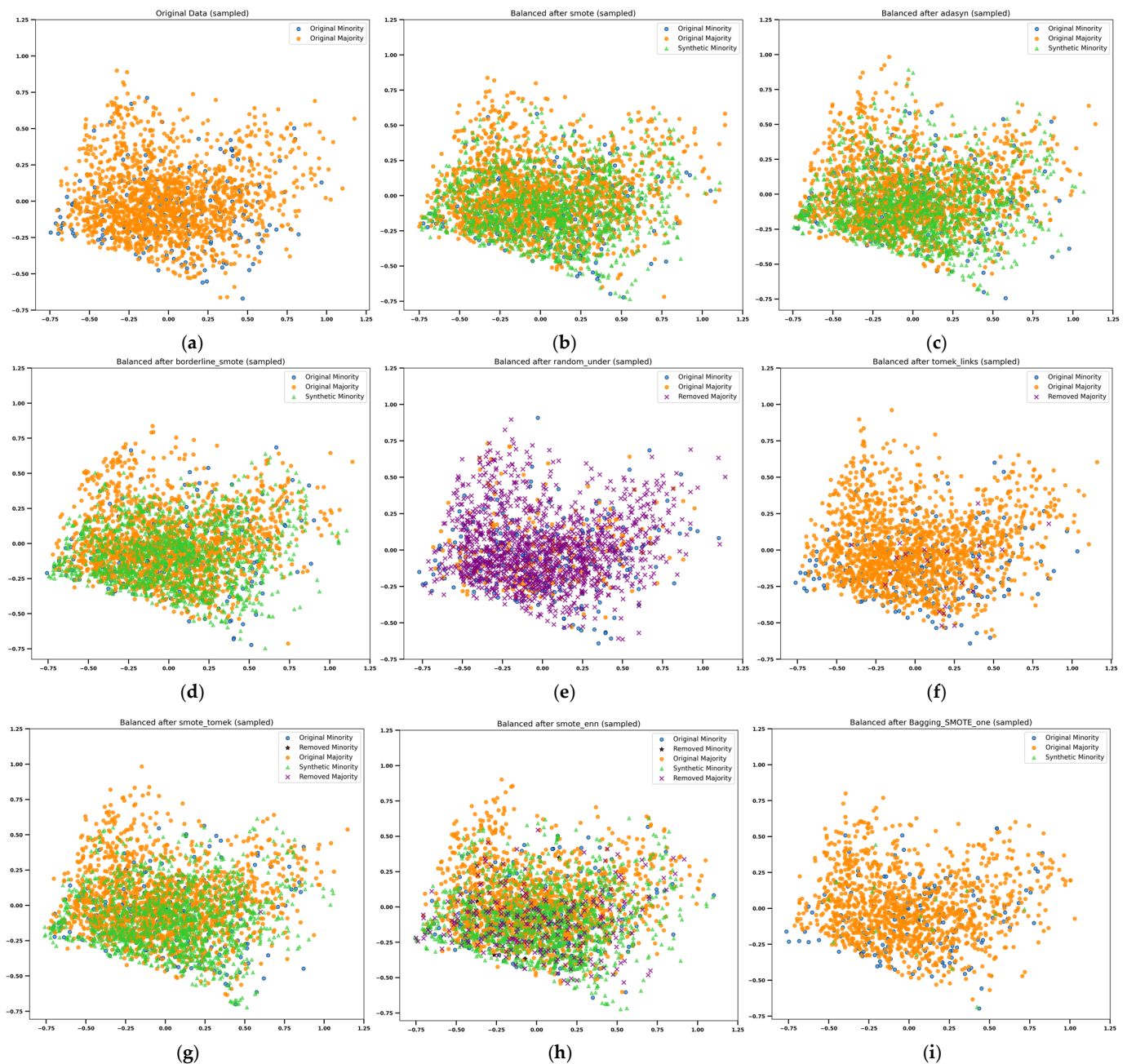


Figure 4. Sample distribution after resampling. (a) Original training data; (b) SMOTE; (c) ADASYN; (d) Borderline-SMOTE; (e) RU; (f) Tomek Links; (g) SMOTE-Tomek; (h) SMOTE-ENN; (i) Bagging-SMOTE. Legends and marker styles are unified: blue circles = original minority, orange dots = original majority, green triangles = synthetic minority, purple crosses = removed majority, red stars = removed minority. Only 10% of samples are shown for clarity.

Oversampling techniques such as SMOTE, ADASYN, and Borderline-SMOTE elevated the number of distressed (minority) samples from 2227 to 16,241, thereby achieving a fully balanced class distribution with a 1:1 ratio. In contrast, undersampling techniques such as RU addressed the imbalance by reducing the number of non-distressed (majority) samples down to 2227, equalizing the class frequencies at the cost of discarding a substantial portion of the majority class data.

Hybrid approaches implemented more nuanced balancing strategies. For instance, Tomek Links applied a light filtering mechanism, resulting in a slight reduction in majority class samples (from 16,241 to 16,021), while keeping the minority class unchanged, maintaining a high imbalance ratio of approximately 7.19:1. This reflects its design, which

aims to eliminate overlapping or borderline instances rather than adjust class frequencies aggressively.

Table 4. Class distribution comparison before and after resampling.

Methods	Class Distribution		
	Class 0	Class 1	Class Ratio (0:1)
Original data	16,241	2227	7.29:1
SMOTE	16,241	16,241	1:1
ADASYN	16,241	16,273	1:1
Borderline-SMOTE	16,241	16,241	1:1
RUS	2227	2227	1:1
Tomek Links	16,021	2227	7.19:1
SMOTE-Tomek	16,229	16,229	1:1
SMOTE-ENN	13,702	16,018	0.86:1
Bagging-SMOTE	16,241	2436	6.67:1

Class 0 denotes non-distressed samples, and Class 1 denotes distressed samples.

Other hybrid techniques, such as SMOTE-Tomek, combined synthetic sample generation with instance filtering to achieve perfect balance, increasing the minority class to 16,229 while slightly reducing the majority class to a comparable number. SMOTE-ENN, which applies to a more aggressive noise-filtering mechanism after oversampling, led to an imbalanced but cleaner dataset, yielding 16,018 distressed and 13,702 non-distressed samples, with a class ratio of 0.86:1. This filtering step aims to remove noisy or ambiguous instances, improving data quality and potentially enhancing model generalization.

Bagging-SMOTE integrates bootstrapped training with SMOTE within an ensemble framework, maintaining consistent resampling outcomes across folds. In each fold of the cross-validation process, four estimators were used for training and one for validation. The majority class in the training estimators comprised 16,241 samples, and SMOTE synthetically increased the minority class samples to about 2436 based on the defined sampling strategy of 0.15. Bagging-SMOTE effectively reshapes the data space by generating synthetic minority instances within the existing data manifold, mitigating class imbalance while maintaining diversity across ensemble folds.

Overall, the visualization and tabulated results highlight the diverse impacts of various resampling techniques on class distribution. These adjustments are critical for addressing the severe class imbalance inherent in financial distress prediction tasks. By balancing the dataset, either through oversampling, undersampling, or hybrid resampling, these techniques enhance the model's capacity to detect distressed samples, which would otherwise be underrepresented and potentially overlooked during the learning process due to the overwhelming dominance of the majority class.

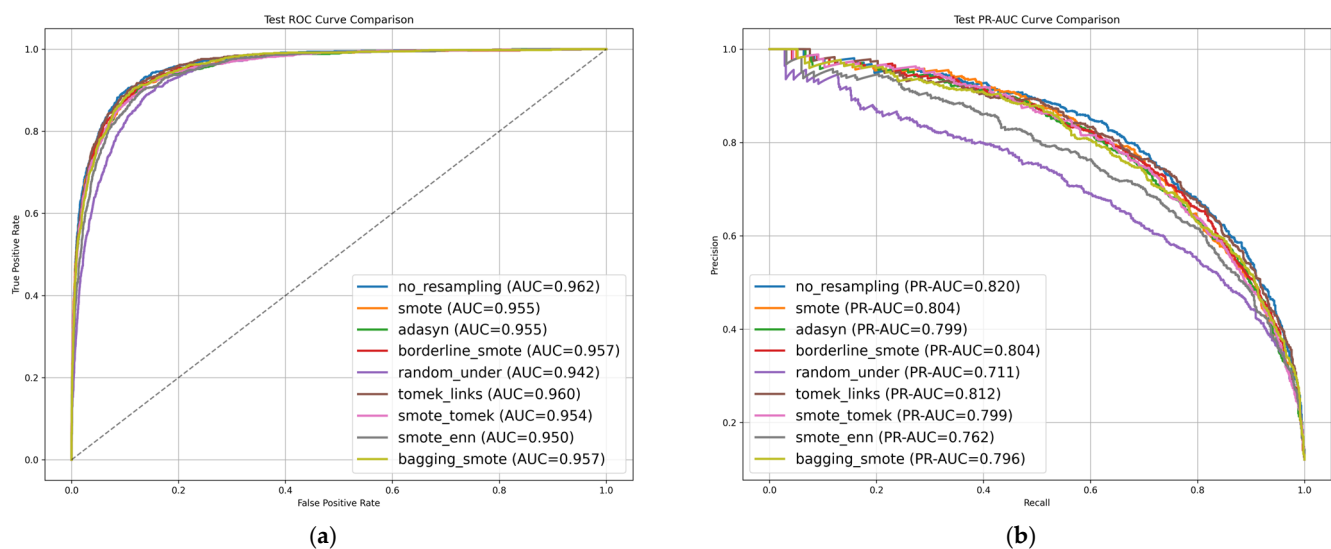
4.2. Impact of Resampling Techniques Toward Prediction Model

To evaluate the effectiveness of resampling techniques in enhancing the model's prediction performance of predicting financial distress, we used the original imbalanced test data to test. Table 5 and Figure 5 summarize the prediction performance of XGBoost models combined with various resampling techniques, including no-resampling, SMOTE, Borderline-SMOTE, ADASYN, SMOTE-ENN, SMOTE-Tomek, Tomek Links, RUS, and Bagging-SMOTE. The examined metrics are accuracy, precision, recall, F1-score, the area under the ROC curve (AUC), the area under the precision-recall curve (PR-AUC), the Matthews Correlation Coefficient (MCC), sampling time, and training time.

Table 5. Classification results on the test set and time costs.

Methods	Accuracy	Precision	Recall	F1	AUC	PR-AUC	MCC	Sampling Time (s)	Training Time (s)
Original data	0.94	0.85	0.60	0.71	0.96	0.82	0.68	-	42.6
SMOTE	0.94	0.77	0.69	0.73	0.96	0.80	0.70	2.4	70.0
ADASYN	0.94	0.76	0.68	0.72	0.96	0.80	0.68	2.3	70.4
Borderline-SMOTE	0.94	0.77	0.69	0.73	0.96	0.80	0.69	2.4	71.4
RUS	0.87	0.48	0.88	0.62	0.94	0.71	0.58	1.6	14.9
Tomek Links	0.94	0.82	0.61	0.70	0.96	0.81	0.68	2.7	41.0
SMOTE-Tomek	0.94	0.75	0.69	0.72	0.95	0.80	0.69	5.9	68.5
SMOTE-ENN	0.92	0.63	0.77	0.69	0.95	0.76	0.65	6.1	58.1
Bagging-SMOTE	0.93	0.74	0.69	0.72	0.96	0.80	0.68	-	400.3

Values in boldface indicate the best result from the metrics.

**Figure 5.** Comparison of ROC and PR-AUC curves. (a) ROC curve; (b) PR-AUC curve.

In the task of financial distress prediction, resampling techniques significantly influence model performance by mediating the trade-off between precision and recall. The benchmark model without any resampling achieves the highest precision (0.85), indicating that most samples it predicts as distressed are indeed distressed. However, its relatively low recall (0.60) indicates that it fails to identify a considerable number of actual distressed cases, revealing a sensitivity issue likely caused by class imbalance. Despite its simplicity and fast processing (training time: 42.6 s), the original model may not be reliable for early warning systems due to under-identification of distressed observations.

In contrast, RUS achieves the highest recall (0.88), demonstrating its effectiveness in detecting distressed samples, but this comes at the cost of significantly reduced precision (0.48) and overall accuracy (0.87). This pattern suggests that while undersampling improves minority class detection, it does so by introducing more false positives. This occurs because RUS balances the classes by aggressively reducing the majority class, resulting in the model learning comprehensive patterns of the minority class but only limited information from the majority class. However, RUS is the most computationally efficient among all methods (training time: 14.9 s), making it suitable for scenarios where detection sensitivity is prioritized and resources are limited. Tomek Links offers a moderate recall of 0.61 and a precision of 0.82, resulting in an F1-score of 0.70 and MCC of 0.68. Its runtime remains low,

requiring 2.7 s for sampling and only 41 s for training, making it a time-efficient option with strong precision.

Among oversampling techniques, SMOTE, ADASYN, and Borderline-SMOTE present a more balanced and stable performance. These techniques improve recall to approximately 0.68–0.69 while maintaining moderate precision levels (0.76–0.77), resulting in relatively strong F1-scores (0.72–0.73). Additionally, their AUC values (0.96) and PR-AUC values (0.80) remain consistently high, indicating strong discriminative power across both classes. MCC ranges from 0.68 to 0.70, further confirming the models' balanced classification ability. Their computational demands are modest, with sampling times around 2.3–2.4 s and training durations of approximately 70 s. These results suggest that synthetic oversampling is effective in alleviating the effects of class imbalance while preserving predictive accuracy, making it a reliable choice for financial distress prediction tasks.

Hybrid resampling techniques offer additional advantages by refining minority class learning and reducing noise in the feature space. SMOTE-Tomek, which combines SMOTE with Tomek Links, yields more balanced performance (recall: 0.69, precision: 0.75, F1-score: 0.72, MCC: 0.69) while slightly increasing computation time (5.9 s sampling, 68.5 s training), which is still acceptable for most financial applications. SMOTE-ENN achieves the highest recall among all oversampling variants (0.77) because its ENN component removes ambiguous or misclassified majority class instances, reducing the majority class to 13,702 samples (see Table 4), while its SMOTE component increases the minority class to 16,018 samples (see Table 4) by generating synthetic examples. This dual process enhances the model's ability to detect minority class cases. However, it may also introduce noise and eliminate some informative majority class samples, leading to more false positives and thus lower precision (0.63). As a result, the F1-score is moderate (0.69), reflecting the trade-off between improved recall and reduced precision. Its sampling and training times are 6.1 s and 58.1 s, respectively, indicating high computational efficiency.

Meanwhile, Bagging-SMOTE demonstrates competitive results across all metrics, achieving balanced recall (0.69), precision (0.74), and F1-score (0.72), supported by AUC (0.96), PR-AUC (0.80), and MCC (0.68). Although its training time is relatively long (400.3 s), the combination of oversampling and ensemble learning significantly enhances generalization performance and stability, especially for highly imbalanced datasets such as those used in financial distress prediction.

Table 6 compares the performance of these resampling techniques on the test set using confusion matrix metrics: True Negatives (TN), False Positives (FP), False Negatives (FN), and True Positives (TP). These outcomes are further illustrated in Figure 6. The results highlight the trade-offs inherent in each method, particularly the balance between enhancing the detection of minority (i.e., TP: Class 1, distressed) cases and preserving the accuracy of majority (i.e., TN: Class 0, non-distressed) classifications.

Using the model trained without any resampling as a baseline, it achieved high accuracy in identifying non-distressed samples (TN = 6859; FP = 102), reflecting a strong bias toward the majority class. However, its performance in detecting distressed cases was notably weaker (TP = 575; FN = 379), highlighting the adverse impact of class imbalance. The resulting low sensitivity undermines the model's practical effectiveness in early financial distress detection, where accurate identification of minority cases is critical.

To mitigate this issue, we compared a series of resampling techniques, including oversampling, undersampling, and hybrid techniques. Among the oversampling methods, SMOTE, ADASYN, and Borderline-SMOTE all balanced the training set to a 1:1 ratio and improved recall for the minority class. SMOTE achieved TP = 661 (FN = 293), ADASYN yielded TP = 653 (FN = 301), and Borderline-SMOTE produced TP = 659 (FN = 295). However, these improvements in sensitivity came at the cost of increased false positives, 198,

210, and 202, respectively, compared to only 102 in the baseline model without resampling. The results suggest that while these methods enhance minority detection, they do so at the cost of greater misclassification of non-distressed instances, thereby reducing overall precision.

Table 6. Confusion matrix on the test set.

Methods	Test Set Class 0: 6961, Class 1: 954			
	TN	FP	FN	TP
Original data	6859	102	379	575
SMOTE	6763	198	293	661
ADASYN	6751	210	301	653
Borderline-SMOTE	6759	202	295	659
RUS	6040	921	117	837
Tomek Links	6833	128	370	584
SMOTE-Tomek	6746	215	295	659
SMOTE-ENN	6528	433	217	737
Bagging-SMOTE	6734	227	291	663

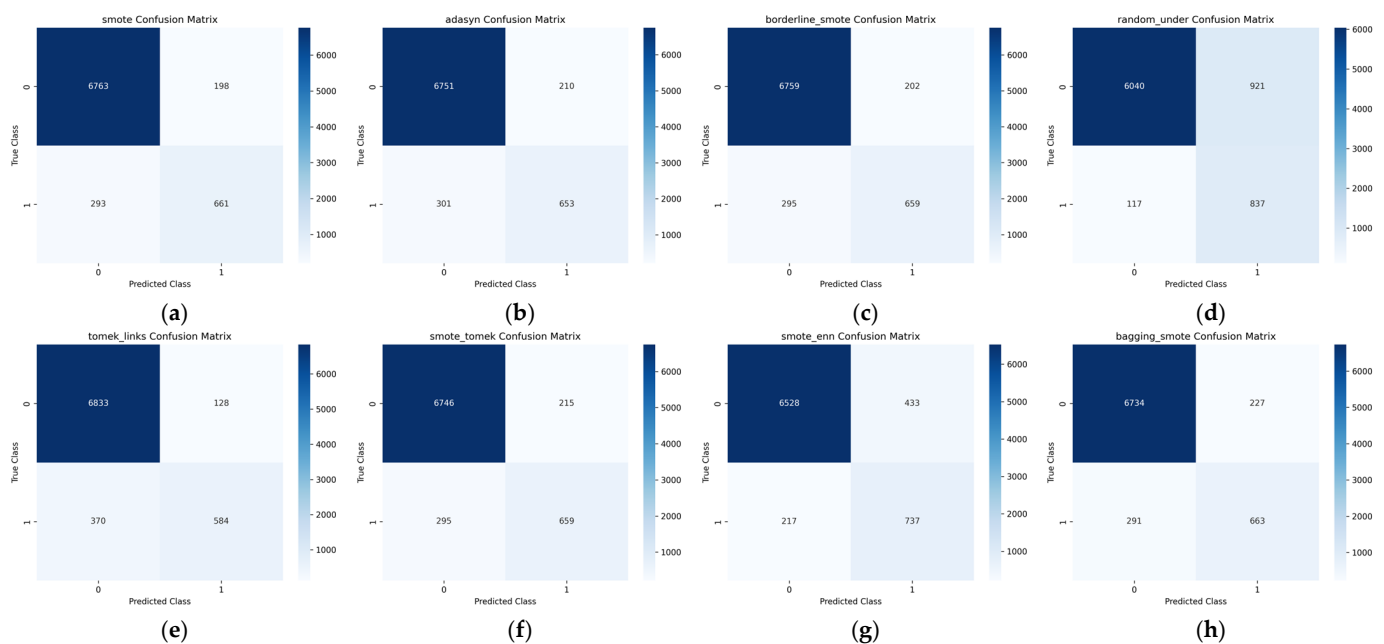


Figure 6. The comparison of the confusion matrix. (a) SMOTE; (b) ADASYN; (c) Borderline-SMOTE; (d) RUS; (e) Tomek Links; (f) SMOTE-Tomek; (g) SMOTE-ENN; (h) Bagging-SMOTE.

In contrast, RUS, which aggressively balanced the classes by reducing the majority to match the minority, achieved the highest recall (TP = 837; FN = 117). However, this came with the highest false positive count (FP = 921) and a reduction in true negatives to 6040, indicating substantial information loss and impaired model precision. A more conservative undersampling approach, Tomek Links, preserved most of the majority data, modestly improved recall (TP = 584), and maintained a relatively low FP (128). This confirms that selective undersampling can improve minority detection without drastically compromising specificity.

Hybrid methods, such as SMOTE-Tomek and SMOTE-ENN, aimed to combine the advantages of both oversampling and data cleaning. SMOTE-Tomek achieved a balanced training set and yielded TP = 659, FN = 295, and FP = 215, offering a better balance between

sensitivity and specificity compared to SMOTE alone. In contrast, SMOTE-ENN produced a higher TP (737) and lower FN (217) but incurred a sharp increase in FP (433), resulting in a significant drop in precision. While SMOTE-ENN demonstrated strong recall, its high false alarm rate makes it less desirable in practical scenarios requiring high precision.

Finally, Bagging-SMOTE demonstrated a strong balance between sensitivity and specificity. It achieved TP = 663, FN = 291, and FP = 227, resulting in a sensitivity of 69.5% and a specificity of 96.8%. Compared to standard SMOTE (TP = 661, FN = 293, FP = 198), Bagging-SMOTE delivered marginal improvements in both sensitivity and specificity, albeit with a slight increase in false positives. The ensemble design effectively stabilizes predictions by averaging across multiple base learners, thereby reducing the variance typically introduced by oversampling. These results suggest that Bagging-SMOTE enhances the model's generalization capability in the context of highly imbalanced financial distress prediction tasks.

5. Discussion

The experimental analysis highlights the crucial role of class imbalance handling in financial distress prediction. In the highly imbalanced dataset used, where distressed samples represent a small minority, XGBoost tended to bias predictions toward the majority (non-distressed) class, resulting in high precision but poor sensitivity to the minority class (such as the model with no sampling). This necessitates the adoption of resampling techniques to mitigate the imbalance and improve minority class detection.

The results revealed substantial performance variability across different resampling techniques, particularly in metrics sensitive to minority class detection, such as recall, precision, and F1-score. Standard SMOTE achieved the highest F1-score (0.73) and the best Matthews Correlation Coefficient (MCC) score (0.70) among all tested techniques. These results indicated that SMOTE was particularly effective in improving the model's ability to detect distressed samples while maintaining a sound balance between precision and recall. The computational demands of synthetic oversampling methods (such as SMOTE and ADASYN) were also modest, with sampling times around 2.3–2.4 s and training durations of approximately 70 s, making them practical for real-world applications. The synthetic generation of minority class samples through interpolation appeared to have successfully generalized the minority class distribution in the feature space, thereby enhancing classifier robustness without introducing significant noise or overfitting.

Among the oversampling and hybrid techniques, Borderline-SMOTE and SMOTE-Tomek consistently delivered more balanced outcomes. For example, compared to standard SMOTE, both techniques achieved similar or slightly improved recall (up to 0.69 for SMOTE-Tomek), highlighting their enhanced ability to identify distressed samples. This suggests that targeted oversampling near decision boundaries (as in Borderline-SMOTE) and the hybrid removal of overlapping instances (as in SMOTE-Tomek) can improve the model's sensitivity to "hard-to-classify" cases while preserving the integrity of the majority class distribution. SMOTE-Tomek offered a good trade-off between recall (0.69), precision (0.75), and computational efficiency (sampling: 5.9 s, training: 68.5 s). These findings reinforce the value of SMOTE as a strong baseline oversampling technique in high-stakes financial prediction tasks while also emphasizing the added benefits of more advanced resampling techniques that focus on decision boundary refinement and noise reduction.

In contrast, Random Undersampling (RUS) achieved the highest recall (0.88) but at the cost of much lower precision (0.48) and overall accuracy (0.87). However, RUS was the most computationally efficient among all methods (training time: 14.9 s, sampling time: 1.6 s), making it suitable for scenarios where detection sensitivity is prioritized and resources are limited. Tomek Links also offered a time-efficient option (sampling: 2.7 s, training: 41 s)

with strong precision (0.82) and moderate recall (0.61), making it attractive for applications where computational speed is critical.

In the Bagging-SMOTE technique, stratified SMOTE was independently applied to each bootstrapped subset of the training data using a sampling ratio of 0.15, and final predictions were obtained through ensemble averaging. While Bagging-SMOTE did not achieve the highest performance in all evaluation metrics, it demonstrated comparable and stable results, including an AUC of 0.96, F1-score of 0.72, and PR-AUC of 0.80. Notably, Bagging-SMOTE maintained a balanced trade-off between precision (0.74) and recall (0.69) and yielded a stable MCC of 0.68, indicating strong generalization capability. However, this came at the cost of a much longer training time (400.3 s), which may limit its practicality in time-sensitive applications. Its performance was particularly notable given the minimal modification of the original dataset, which preserved data integrity while still improving minority class detection. These results underscore the practical effectiveness of combining stratified oversampling at an appropriate sampling ratio with ensemble learning, particularly in enhancing the detection of minority-class (distressed) cases while mitigating model variance, a trade-off that is especially valuable in real-world financial scenarios where both predictive accuracy and data fidelity are critical.

In real-world financial distress prediction, the costs associated with false positives (incorrectly flagging a healthy company as distressed) and false negatives (failing to identify an actually distressed company) are highly asymmetric and context dependent. False negatives are often more detrimental, as overlooking a distressed firm can lead to significant financial losses, regulatory penalties, or reputational harm. In contrast, false positives may result in unnecessary interventions, increased monitoring costs, or missed business opportunities. Therefore, the choice of resampling technique should go beyond overall accuracy and consider the specific cost structure, business objectives, and risk tolerance of the application.

For example, techniques that emphasize recall, such as Borderline-SMOTE or SMOTE-Tomek, may be preferred in early warning systems, where minimizing missed distress cases outweighs the cost of false alarms. Conversely, when operational efficiency and resource management are paramount, methods such as Bagging-SMOTE, which provide a more balanced trade-off between precision and recall, may be more appropriate. Additionally, when computational resources or time constraints are a concern, RUS or Tomek Links may be prioritized due to their superior efficiency, despite their trade-offs in predictive performance.

Overall, the experimental findings highlight the importance of selecting resampling techniques that are closely aligned with the specific objectives and misclassification costs of the application to maximize the real-world value of financial distress prediction models.

Future research may extend this work by investigating uncertainty-aware sampling frameworks, where sample selection is guided by measures of predictive confidence or entropy. Additionally, incorporating explainable machine learning techniques, such as SHAP, to assess the influence of synthetic samples on model behavior could provide valuable insights for refining the resampling-model feedback loop. These advancements hold potential to support the development of more intelligent, interpretable, and effective resampling strategies, particularly critical in high-stakes domains such as financial risk assessment and corporate distress prediction.

6. Conclusions

In this study, we systematically compared eight resampling techniques to address the class imbalance problem in financial distress prediction, using a real-world dataset comprising 26,383 firm-quarter samples from 639 Chinese A-share listed companies spanning

the years 2007 to 2024. With only 12.1% of the samples labeled as distressed, the dataset presents a typical yet challenging imbalanced classification task. To ensure consistency and fair evaluation, the XGBoost algorithm was employed as the base classifier across all techniques, with identical model structures and hyperparameter settings.

The experimental results show that resampling techniques have a significant impact on model performance. This effect is especially notable in metrics sensitive to minority-class detection, such as precision, recall, F1-score, and Matthews Correlation Coefficient (MCC). Several key findings emerged from the comparative analysis:

- Resampling is critical for improving minority class detection. Traditional oversampling methods, especially SMOTE, showed meaningful improvements in F1-score (up to 0.73) and MCC (up to 0.70). These results indicate enhanced model robustness without excessive overfitting. Hybrid techniques such as SMOTE-Tomek and Borderline-SMOTE further improved recall while almost maintaining precision. This highlights their value in refining decision boundaries and reducing class overlap. Importantly, these methods also demonstrated modest computational demands, with sampling times around 2–6 s and training times of approximately 70 s. Therefore, they are practical for real-world applications.
- Trade-offs between recall and precision, as well as computational efficiency, must be context aware. Aggressive techniques such as RUS yielded the highest recall (0.85). However, this came at the cost of poor precision (0.46) and diminished generalization performance, mainly due to the loss of informative majority-class samples. RUS was also the most computationally efficient method, with a training time of 14.9 s and a sampling time of 1.6 s. This makes it suitable for scenarios where detection sensitivity and speed are prioritized. In contrast, Tomek Links offered a strong balance between precision (0.82) and efficiency (sampling: 2.7 s, training: 41 s). This makes it attractive for applications where computational speed and minimizing false positives are critical. SMOTE-ENN achieved a high recall (0.77) but at a moderate precision level (0.63), suggesting a tendency toward increased false positives. Its computational cost was also reasonable (sampling: 6.1 s, training: 58.1 s).
- Bagging combined with stratified SMOTE (Bagging-SMOTE) achieved a strong overall balance across evaluation metrics. Applying stratified SMOTE with an appropriate sampling ratio (0.15) to each bootstrap sample and aggregating predictions through bagging led to competitive results. These included AUC of 0.96, F1-score of 0.72, PR-AUC of 0.80, and MCC of 0.68. However, this method required much longer training time (400.3 s), which may limit its practicality in time-sensitive applications. These results show that ensemble-based resampling can improve both sensitivity and generalization, which is important in financial applications.
- Resampling techniques selection should be tailored to application goals. For early warning systems that prioritize recall and timely identification of financially distressed companies, techniques such as Bagging-SMOTE, Borderline-SMOTE, SMOTE-Tomek, and SMOTE are recommended. In contrast, when minimizing false positives is important, such as in regulatory flagging or risk-adjusted decision-making, more conservative methods, such as Tomek Links, may be more appropriate.

Practitioners should carefully consider the specific cost structure and operational priorities of their financial distress prediction tasks. For applications where missing a distressed firm is particularly costly, we recommend prioritizing resampling methods that maximize recall, such as Bagging-SMOTE or Borderline-SMOTE. Conversely, in scenarios where false alarms are more problematic, more conservative techniques such as Tomek Links may be preferable. Additionally, when computational resources or time constraints are a concern, RUS and Tomek Links offer practical alternatives due to their superior

efficiency, despite their trade-offs in predictive performance. Ensemble-based approaches can offer a robust balance between sensitivity and specificity, making them suitable for high-stakes financial environments.

Future research could advance these findings by integrating uncertainty-aware sampling strategies, where sample selection is driven by predictive confidence. Leveraging explainable machine learning techniques could further improve the synergy between re-sampling and modeling, while enhancing the transparency of interpretability. Additionally, extending these techniques to datasets from diverse markets, such as the U.S. or Europe, would provide valuable insights into their generalizability, making a valuable avenue for future exploration. Researchers could also investigate advanced class imbalance handling methods, such as adaptive resampling or hybrid approaches, combined with cost-sensitive learning. Finally, real-world case studies are needed to evaluate the practical impact and deployment of these techniques in financial risk assessment and early warning systems.

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References

1. Song, Y.; Jiang, M.; Li, S.; Zhao, S. Class-imbalanced Financial Distress Prediction with Machine Learning: Incorporating Financial, Management, Textual, and Social Responsibility Features into Index System. *J. Forecast.* **2024**, *43*, 593–614. [\[CrossRef\]](#)
2. Engin, U. Financial distress prediction from time series data using xgboost: Bist100 of borsa istanbul. *Doğuş Üniversitesi Derg.* **2023**, *24*, 589–604. [\[CrossRef\]](#)
3. Chawla, N.V.; Bowyer, K.W.; Hall, L.O.; Kegelmeyer, W.P. SMOTE: Synthetic Minority Over-Sampling Technique. *J. Artif. Intell. Res.* **2002**, *16*, 321–357. [\[CrossRef\]](#)
4. Wang, B.; Mei, C.; Wang, Y.; Zhou, Y.; Cheng, M.-T.; Zheng, C.-H.; Wang, L.; Zhang, J.; Chen, P.; Xiong, Y. Imbalance Data Processing Strategy for Protein Interaction Sites Prediction. *IEEE/ACM Trans. Comput. Biol. Bioinf.* **2021**, *18*, 985–994. [\[CrossRef\]](#)
5. Cai, T. Breast Cancer Diagnosis Using Imbalanced Learning and Ensemble Method. *Appl. Comput. Math.* **2018**, *7*, 146. [\[CrossRef\]](#)
6. Elreedy, D.; Atiya, A.F. A Novel Distribution Analysis for SMOTE Oversampling Method in Handling Class Imbalance. In *Computational Science—Proceedings of the ICCS 2019: 19th International Conference, Faro, Portugal, 12–14 June 2019*; Lecture Notes in Computer Science; Springer International Publishing: Cham, Switzerland, 2019; pp. 236–248. [\[CrossRef\]](#)
7. Alex, S.A.; Nayahi, J.J.V. Classification of Imbalanced Data Using SMOTE and AutoEncoder Based Deep Convolutional Neural Network. *Int. J. Unc. Fuzz. Knowl. Based Syst.* **2023**, *31*, 437–469. [\[CrossRef\]](#)
8. Han, H.; Wang, W.-Y.; Mao, B.-H. Borderline-SMOTE: A New Over-Sampling Method in Imbalanced Data Sets Learning. In *Advances in Intelligent Computing*; Springer Berlin Heidelberg: Berlin, Heidelberg, 2005; Volume 3644, pp. 878–887. ISBN 978-3-540-28226-6. [\[CrossRef\]](#)
9. Chen, C.; Shen, W.; Yang, C.; Fan, W.; Liu, X.; Li, Y. A New Safe-Level Enabled Borderline-SMOTE for Condition Recognition of Imbalanced Dataset. *IEEE Trans. Instrum. Meas.* **2023**, *72*, 1–10. [\[CrossRef\]](#)
10. Glazkova, A. A Comparison of Synthetic Oversampling Methods for Multi-Class Text Classification. *arXiv* **2020**, arXiv:2008.04636. [\[CrossRef\]](#)

11. He, H.; Bai, Y.; Garcia, E.A.; Li, S. ADASYN: Adaptive Synthetic Sampling Approach for Imbalanced Learning. In Proceedings of the 2008 IEEE International Joint Conference on Neural Networks (IEEE World Congress on Computational Intelligence), Hong Kong, China, 1–8 June 2008; IEEE: Hong Kong, China, 2008; pp. 1322–1328. [\[CrossRef\]](#)
12. Pristyanto, Y.; Nugraha, A.F.; Dahlan, A.; Wirasakti, L.A.; Ahmad Zein, A.; Pratama, I. Multiclass Imbalanced Handling Using ADASYN Oversampling and Stacking Algorithm. In Proceedings of the 2022 16th International Conference on Ubiquitous Information Management and Communication (IMCOM), Seoul, Republic of Korea, 3 January 2022; IEEE: Seoul, Republic of Korea, 2022; pp. 1–5. [\[CrossRef\]](#)
13. Arifin, M.A.S.; Stiawan, D.; Yudho Suprpto, B.; Susanto, S.; Salim, T.; Idris, M.Y.; Budiarto, R. Oversampling and Undersampling for Intrusion Detection System in the Supervisory Control and Data Acquisition IEC 60870-5-104. *IET Cyber-Phys. Syst. Theory Appl.* **2024**, *9*, 282–292. [\[CrossRef\]](#)
14. Batista, G.E.A.P.A.; Prati, R.C.; Monard, M.C. A Study of the Behavior of Several Methods for Balancing Machine Learning Training Data. *ACM SIGKDD Explor. Newsl.* **2004**, *6*, 20–29. [\[CrossRef\]](#)
15. Kaushik, M.M.; Mahmud, S.M.H.; Kabir, M.A.; Nandi, D. The Effects of Class Rebalancing Techniques on Ensemble Classifiers on Credit Card Fraud Detection: An Empirical Study. In Proceedings of the Applied Data Science and Smart Systems, Rajpura, India, 4–5 November 2022; AIP Publishing LLC: Rajpura, India, 2023; p. 030011. [\[CrossRef\]](#)
16. Zang, J.; Li, H. Abnormal Traffic Detection Based on Data Augmentation and Hybrid Neural Network. In Proceedings of the 2024 2nd International Conference on Signal Processing and Intelligent Computing (SPIC), Guangzhou, China, 20 September 2024; IEEE: Guangzhou, China, 2024; pp. 249–253. [\[CrossRef\]](#)
17. Putra, L.G.R.; Marzuki, K.; Hairani, H. Correlation-Based Feature Selection and Smote-Tomek Link to Improve the Performance of Machine Learning Methods on Cancer Disease Prediction. *Eng. Appl. Sci. Res.* **2023**, *50*, 577583. [\[CrossRef\]](#)
18. Yang, F.; Wang, K.; Sun, L.; Zhai, M.; Song, J.; Wang, H. A Hybrid Sampling Algorithm Combining Synthetic Minority Over-Sampling Technique and Edited Nearest Neighbor for Missed Abortion Diagnosis. *BMC Med. Inform. Decis. Mak.* **2022**, *22*, 344. [\[CrossRef\]](#)
19. Wang, W.; Liang, Z. Financial Distress Early Warning for Chinese Enterprises from a Systemic Risk Perspective: Based on the Adaptive Weighted XGBoost-Bagging Model. *Systems* **2024**, *12*, 65. [\[CrossRef\]](#)
20. Liu, W.; Fan, H.; Xia, M.; Pang, C. Predicting and Interpreting Financial Distress Using a Weighted Boosted Tree-Based Tree. *Eng. Appl. Artif. Intell.* **2022**, *116*, 105466. [\[CrossRef\]](#)
21. Wu, C.; Chen, X.; Jiang, Y. Financial Distress Prediction Based on Ensemble Feature Selection and Improved Stacking Algorithm. *Kybernetes*, 2024; *ahead-of-print*. [\[CrossRef\]](#)
22. Seiffert, C.; Khoshgoftaar, T.M.; Van Hulse, J.; Napolitano, A. RUSBoost: A Hybrid Approach to Alleviating Class Imbalance. *IEEE Trans. Syst. Man Cybern. A* **2010**, *40*, 185–197. [\[CrossRef\]](#)
23. Díez López, C.; Montiel González, D.; Vidaki, A.; Kayser, M. Prediction of Smoking Habits From Class-Imbalanced Saliva Microbiome Data Using Data Augmentation and Machine Learning. *Front. Microbiol.* **2022**, *13*, 886201. [\[CrossRef\]](#)
24. Shaikh, S.; Daudpota, S.M.; Imran, A.S.; Kastrati, Z. Towards Improved Classification Accuracy on Highly Imbalanced Text Dataset Using Deep Neural Language Models. *Appl. Sci.* **2021**, *11*, 869. [\[CrossRef\]](#)
25. Kotb, M.H.; Ming, R. Comparing SMOTE Family Techniques in Predicting Insurance Premium Defaulting Using Machine Learning Models. *Int. J. Adv. Comput. Sci. Appl.* **2021**, *12*, 621–629. [\[CrossRef\]](#)
26. Hairani, H.; Anggrawan, A.; Priyanto, D. Improvement Performance of the Random Forest Method on Unbalanced Diabetes Data Classification Using Smote-Tomek Link. *JOIV Int. J. Inform. Vis.* **2023**, *7*, 258. [\[CrossRef\]](#)

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